MUTUAL INDUCTANCE OF PARALLEL TRANSMISSION LINES   file: mline.mcd

Formulas included in this spreadsheet:

Mutual inductance of two lines   MLINE()

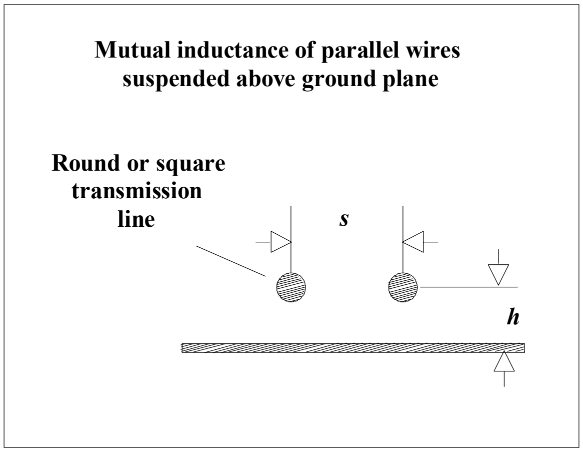
Variables used:

s   Separation between wire centers (in.)

h   Height of wires above ground (in.)

x   Length of parallel span (in.)

(We assume that two identical transmission lines share a parallel run of length x, with a horizontal separation s.)



mline

Let L equal the inductance (H) of the first transmission line of length x (use formula for round, microstrip, or stripline geometry as appropriate):

$$MLINE(L,s,h) := L\cdot\left[\frac{1}{1+\left(\frac{s}{h}\right)^2}\right]$$